

Modelling the ion exchange process in glass: Phenomenological approaches and perspectives

A. Quaranta^{a,*}, E. Cattaruzza^b, F. Gonella^b

^a *Dipartimento di Ingegneria dei Materiali e delle Tecnologie Industriali (DIMTI), Università di Trento, via Mesiano 77, I-38050 Povo, Trento, Italy*

^b *Dipartimento di Chimica Fisica, Università Ca' Foscari Venezia, Dorsoduro 2137, I-30123 Venezia, Italy*

Received 17 October 2007; received in revised form 7 November 2007; accepted 21 November 2007

Abstract

A great deal of work has been devoted in the last decades to the study of the ion exchange process modelling, in order to tailor the physical properties of the modified layers, e.g. refractive index or stress profiles. The first models were based on simple ionic diffusion equations, while in the last years it became clear that the glass undergoes local restructuring during the ion exchange even at temperatures lower than the glass transition one. The aim of this work is to review the approaches that lead to the introduction of self-diffusion coefficients depending on the concentration of the migrating cations, and to outline a physical approach in selecting a phenomenological expression for the coefficients. In particular, this approach refers to models developed for describing the ionic diffusion in mixed-alkali glasses.

© 2007 Elsevier B.V. All rights reserved.

Keywords: Theory of diffusion; Ions; Glass

1. Introduction

Ion exchange in silicate glasses have been the subject of much scientific interest since the first applications for surface strengthening [1]. To date it is studied for the fabrication of passive and active optical waveguides [2–5], of stress modulated systems [6], for the control of the microstructural properties of glasses [7] and for deepening the understanding of different types of environment–glass interactions [8]. Usually the process is performed by immersing the glass substrate in a molten salt bath containing the required metallic ion, which exchanges with the alkali ion species of the glass giving rise to a diffusion profile whose shape and depth depend on the ion diffusion coefficients and on the temperature. In some cases, an external electric field may be applied to control more accurately the profile shape. In other approaches, the diffusion of metallic ions is carried out starting from a metallic film deposited on the glass surface; in this case, the migration is promoted by an external voltage applied between the two faces of the glass slide.

The diffusion of the ionic species in the glass network is described by means of the Nernst–Planck equation for ionic diffusion in electrolytes, which is the simplest and well-established approach. Nonetheless, a more detailed description of the concentration profiles of the doping ion indicates that in most cases the ion exchange equations must take into account diffusion parameters depending on the species concentrations. This is supported by experimental evidences revealing significant structural changes of the ion sites of the glass network after the exchange.

Diffusion models where the transport parameters depend on the species concentration can refer to the models developed for describing the mixed-alkali effect (MAE), where the nonlinear relationship of the concentration with several physical properties, like the electrical conductivity and the ion diffusivity, has been related to the formation of different kind of ion sites into the glass network.

In this work, the description of the ion exchange in silicate glasses is reviewed, highlighting some well-established points and outlining the approaches capable to give a physical meaningful expression for the diffusion coefficients dependent on the concentration. In particular, we will refer to description of the ion exchange process based on the phenomenological Nernst–Planck inter-diffusion equation. This approach is distinct from more complex advances, like molecular dynamics-

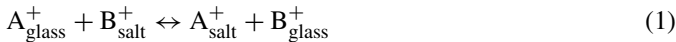
* Corresponding author. Tel.: +39 0461 882450; fax: +39 0461 881945.
E-mail address: quaranta@ing.unitn.it (A. Quaranta).

based models, which are less practical for treatment and engineering applications even if they can give important information on the effects of the different ion species on the glass short-range structure.

2. Ion exchange and the inter-diffusion equation

Ion exchange in silicate glass has been performed up to now mostly by substituting Na^+ ions in sodium silicate and soda-lime glasses where the Na_2O molar concentration was higher than 10%, such to let a sufficiently high amount of doping ions to enter the glass, and thus having the possibility to tailor the surface physical properties. Substitutive ions were in most cases K^+ , for surface strengthening and optical purposes, and Ag^+ for the production of waveguides. In few cases other alkali ions have been used, like Li^+ [9] and Cs^+ [10,11], and $\text{Cu}^+/\text{Cu}^{2+}$ [12–14], Tl^+ [15], Er^{3+} [16] and more recently Co^{2+} and Au^{3+} [17,18] for the production of passive and/or active optical materials. The treatment procedure is usually based on the immersion of the glass slides in molten salt baths, where an external electric field can be applied, at temperatures lower than the glass T_g . In some cases, the ion source is a metallic film deposited on the glass slide, through which an external voltage is applied ionizing and driving the metal ions from the film.

Concerning the salt bath ion exchange, at the glass surface the following chemical balance occurs:



where A^+ is the alkali ion of the glass and B^+ is the doping ion from the salt. The equilibrium at the surface is attained when the electrochemical potentials of the ion species are the same in the two phases. Usually, it is assumed that the surface equilibrium concentration is reached within times much shorter than the time required for the inner diffusion; for this reason, in the analysis of the inter-diffusion process the surface concentration may be kept constant.

In the glass network, the two ion species diffuse following the Nernst–Planck equations for the ion flux of a species i , J_i :

$$J_i = -D_i \left(\frac{\partial \ln a_i}{\partial \ln X_i} \right) \frac{\partial c_i}{\partial x} + u_i E c_i \quad (2)$$

where D_i is the self-diffusion coefficient, a_i the activity of the ion, X_i and c_i the atomic fraction and the concentration of the ion species, respectively, E the local electric field balancing the local charge density when two ions with different mobility are diffusing, and u_i is the ion mobility, related to the diffusion coefficient by the Nernst–Einstein relation:

$$D_i = f_i \frac{kT}{Z_i e} u_i \quad (3)$$

where the factor f_i accounts for the difference between thermal and field-assisted migration, k the Boltzmann constant, T the absolute temperature, Z_i the ion valency, and e is the elementary charge (for an exhaustive description of the term f_i , see Refs. [19,20]). The self-diffusion coefficients follow an Arrhenius

dependence on the temperature:

$$D_i = D_{0i} e^{-\Delta E/kT} \quad (4)$$

where ΔE is the energy barrier for a jump between two adjacent sites and D_{0i} is the maximum diffusion coefficient (at infinite temperature). The first and most known expression for the energy barrier is due to Anderson and Stuart [21], who assumed it made by two terms, related to the electrostatic interaction with the neighbors and to the elastic strain of the network, treated as a continuum.

The final differential equation for the concentration of the ion is obtained by removing the local electric field term, under the conditions that the local charge neutrality is attained:

$$Z_A J_A + Z_B J_B = 0, \quad Z_A c_A + Z_B c_B = Z_A c_{0A} \quad (5)$$

where c_{0A} is the pristine concentration of A^+ ions in the glass. In the literature [22,23], the correctness of this assumption was verified by explicitly calculating the local electric field, finding that the local charge neutrality is satisfied within 10^{-3} s from the starting of the inter-diffusion process.

By combining Eqs. (1) and (5), the differential equation for the concentration profile $c_B(x, t)$ of a monovalent ion takes the form:

$$\begin{aligned} \frac{\partial c_B}{\partial t} &= -\frac{\partial J_B}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial \ln a_B}{\partial \ln X_B} \frac{D_B D_A}{X_A D_A + X_B D_B} \frac{\partial c_B}{\partial x} \right) \\ &= \frac{\partial}{\partial x} \left(\frac{\partial \ln a_B}{\partial \ln X_B} \tilde{D} \frac{\partial c_B}{\partial x} \right) \end{aligned} \quad (6)$$

where $\tilde{D}$ is also named inter-diffusion coefficient.

This relationship, firstly introduced by Doremus [24], is the equation used for the analysis of the diffusion process of the doping ion in the glass matrix. The term containing the thermodynamic activity is usually neglected or kept constant, following for instance the glass–salt equilibrium studies of Garfinkel [25], who interpolated the experimental data by supposing that the logarithmic derivative is equal to a constant n between 1 and 2. So, the equation can be solved numerically or by means of analytical procedures, by taking into account the dependence on the concentration of the inter-diffusion coefficient. Nevertheless, even this equation, containing a diffusion parameter intrinsically dependent on the ion concentration, can be inadequate to accurately reconstruct the experimental diffusion profiles. This problem can be relevant especially in the case of optical devices, where the number and the values of the effective refractive indices of the guided modes strongly depend on the profile shape of the refractive index [26,27]. For this reason, in many cases it was assumed that the self-diffusion coefficients themselves are not constant during the whole exchange process, but they depend on the cation concentration. The confirmation of this hypothesis came from experimental results evidencing that the short-range structure around the ion sites change after the substitution, in terms of both the local cation coordination number (CN) and the cation–oxygen bond length.

3. Glass structure and ion exchange

For several years, literature reports concerning the process used to consider ion exchange as a mere substitution of the new ion in the Na^+ site, with the possible formation of stress states given by the difference between the ionic radii. With the rising of experimental technique, such as the extended X-ray absorption fine structure (EXAFS) spectroscopy, the knowledge of the short-range glass structure reached new aims. In 1981 Greaves et al. [28] found for the first time that in sodium silicate and soda-lime glasses the coordination of sodium follows a short-range order similar to crystalline silicates, with coordination numbers (CNs) of about 5 for sodium silicate and of about 2 for soda-lime glasses, and with Na–O bond lengths in the range of 0.220–0.240 nm. Moreover, in glasses usually employed for ion exchange experiments, the sodium concentration is so high to form some sort of connected structure in the silica network along percolation channels. This frame, which strongly affects the diffusion mechanism, can change in a more random distribution for the presence of Al_2O_3 , which lowers the number of non-bridging oxygens (NBOs).

Concerning ion exchange, a good deal of work has been devoted to EXAFS analyses on silicate glasses treated by Ag^+ – Na^+ ion exchange. In particular, both Dubiel et al. [29,30] and D'Acapito et al. [31] found that the ionic site around Ag^+ ions is rearranged owing to the formation of nearly covalent bonds with NBOs, leading to the lowering of the Ag–O distance to about 0.215 nm and to the change of the CN from 5 to 3 for low Ag^+ concentrations (12% of exchange) and to 2 for high concentrations (80% of exchange). In particular, the latter structure resembles that of Ag_2O , even if the bond length of this oxide is lower (about 0.204 nm). Moreover, experimental findings and molecular dynamics calculations [32] established that the Ag–O bond length has a slight tendency to decrease with the increasing of the silver content, while an Ag–Ag correlation has been observed attributed to homoatomic interactions between closed d^{10} shells. It is worth noting that the Ag–O bond length has been well reproduced in two different types of glasses, a sodium silicate glass for Dubiel et al. [29,30] and a soda-lime glass for D'Acapito et al. [31].

More recently, it has been found that the Ag–O bond length for low exchange times (few minutes) is higher [33] with respect to what found by Dubiel, who examined glass samples treated for very long times (several hours). This result evidenced a further aspect of the site changes, that is a structural relaxation behavior similar to what observed for the K^+ – Na^+ ion exchange, where the surface stress state tend to relax by increasing the treatment temperature and time [34].

The strong rearrangement of the glass short-range structure around the Ag^+ site explains the absence of relevant stress states, as observed by measuring the birefringence of optical waveguides [35,36]. Indeed, the Ag^+ ionic radius (0.126 nm) is higher than Na^+ (0.095 nm), and could give rise to measurable stress states in the glass network, as occurs for instance in the case of the K^+ – Na^+ ion exchange, where the K^+ ionic radius is 0.133 nm.

Even stronger effects on the glass structure have been evidenced by EXAFS measurements in Cu^+ – Na^+ exchanged

soda-lime glasses [37,38]. In this case, as discussed in the following, the doping ion species diffused into the glass under two oxidation states, Cu^+ and Cu^{2+} , with different coordination geometries. In particular, it has been evidenced a square local symmetry around Cu^{2+} ions and a linear coordination of O atoms around Cu^+ . The bond lengths were characteristic of CuO (around 0.195 nm) where a higher amount of Cu^{2+} was present, and of Cu_2O (around 0.185 nm) where Cu^+ was the predominant species [38]. This strong effect on the structure can be also related to the high temperatures (between 520 and 560 °C) required for melting copper salt eutectics.

Even if EXAFS remain the most important technique in evaluating the local structure of ions in glasses, other kind of analyses have underlined microstructural changes of the glass network upon ion exchange. For instance, photoluminescence measurements evidenced the change of the local arrangement of silver and copper sites with the species concentration, up to the formation of seeds for the growth of metallic clusters in silver exchanged glasses [39–41]. Finally, nuclear magnetic resonance (NMR) measurements on silver and copper exchanged glasses [42–45] indicated a change in the distribution of NBOs, with the production of Q^2 from Q^3 units, without Q^4 formation [44]. A similar behavior has been assumed in K^+ – Na^+ ion exchanged glasses from the analysis of infrared spectra [46]. The effect on the structure lowers with the insertion in the network of Al_2O_3 [45], which is known to increase the network connectivity by decreasing the NBO number. Moreover, Yano et al. [44] pointed out a decrease of the T_g of ion exchanged glasses, reaching temperatures even lower than the exchange temperature for glasses treated for long times.

Other physical properties of the exchanged surface layers are less affected by the changes of the ion site structure. For instance, it has been demonstrated that in Ag^+ or Cu^+ exchanged glasses the increase of the refractive index can be quite well reproduced by a Lorenz–Lorentz dependence on the ion concentration, where the ion polarizability of the doping ion is simply substituted for that of the replaced sodium ions [12,26].

Summarizing, the results reported in the literature outlined that, during ion exchange at temperature lower than the T_g of the pristine glass, both the cation CN and the NBO distribution can change by means of bond breaking and rearrangements of oxygen ions. These findings lead to the conclusion that the diffusion parameters actually depend on the concentration of the two inter-diffusing ion species. Moreover, any model with self-diffusion coefficients depending on the concentration must take into account the modification of the cation sites during the migration.

4. Mixed mobile ion effect on ion exchange

The first detailed analysis on the dependence of self-diffusion coefficients on the concentration of the migrating species can be found in the study of the K^+ – Na^+ ion exchange [47,48]. In this case, a good agreement was found between the inter-diffusion coefficient $\tilde{D}(c)$ calculated with the Boltzmann–Matano [49] numerical procedure and the same calculated by inserting in Eq. (6) the self-diffusion coefficients of mixed-alkali glasses

at different compositions. The residual deviations have been reconciled with the strains produced around a cation jump site during ion exchange and, in turn, with the macroscopic stresses in the glass body. This result suggested that the mixed-alkali effect (MAE) could give a reasonable physical expression of the self-diffusion coefficients during the ion exchange.

The MAE is one of the most widely studied phenomena in glass science. From the point of view of the ion diffusion process, it consists in the lowering of the self-diffusion coefficient of an alkali ion when another species is introduced into the network. Moreover, the self-diffusion coefficient curves of the two alkalis intersect at some particular alkali ratio where both cations have the same mobility. Even if a complete explanation of the MAE is still to come, it has been definitely established that it is related to the stabilization of different site environments for the two alkali species, and to the formation of percolation pathways for the majority ion. This situation resembles that described for ion exchange in the latest paragraph.

In the MAE, the ion sites are fixed in the glass network, and the dependence of the migration parameters on the alkali concentration is related to the modification of the activation energy for the jump between two adjacent sites. In fact, the energy barrier is given by two terms, one accounting for the strain of the network doorway during the jump and another one accounting for the electrostatic interaction energy between neighboring ions. In particular, this latter term could explain the observed dependence on the cation concentration. These terms have been reported for the first time in 1954 by Anderson and Stuart [21] and, with some more sophisticated revision, their expression was substantially confirmed more recently [50,51].

The ion exchange process is in some way more complex, since it involves changes of oxygen bonds, rearrangements of bond angles and, in cases such that of Ag^+ , the formation of bonds with more covalent oxygens with respect to alkali ions. For all these reasons, the dependence of the self-diffusion coefficients could not follow exactly the MAE model, as proposed for example by Ingram et al. [46] in their dynamic structure model for the $\text{K}^+ - \text{Na}^+$ ion exchange.

Even in the light of the concerns expressed above, a model where the concentration of the cationic species affects the diffusion activation energies is physically meaningful if it takes into account the local rearrangement occurring in the glass network by means of concentration-dependent diffusion activation energies. From this point of view, the activation energy can be expressed in a more general way, like that proposed by other authors for MAE [52,53]. For instance, Tomozawa [53] gave the following relationships for mixed-alkali glasses:

$$\begin{aligned}\Delta\epsilon_A &= \Delta\epsilon_{A0} - z f_{AA} \epsilon_{AA} - z f_{AB} \epsilon_{AB}, \\ \Delta\epsilon_B &= \Delta\epsilon_{B0} - z f_{BB} \epsilon_{BB} - z f_{BA} \epsilon_{BA}\end{aligned}\quad (7)$$

where $\Delta\epsilon_{i0}$ is a constant term depending on the network species which are not involved in the diffusion process, z the CN, f_{ij} the probability to find a j -ion near a i -ion and ϵ_{ij} is the interaction energy between an i and a j ion. In particular, for a random distribution of ions, f_{ij} is simply X_j . In this general expression, the interaction energies can assume a deeper meaning, since they

are related also to the tendency of an ion species to be surrounded by similar or dissimilar ions, when either ϵ_{ii} or ϵ_{ij} takes the lower value, respectively.

Up to now, evaluations of the jump energy values has been attempted by Tomozawa starting from mixing enthalpy values and by Greaves from conductivity and EXAFS measurements [54,55]. Moreover, Araujo found an expression, obtained from the statistical thermodynamics relationship of the electrochemical potentials, for evaluating the interaction energy between unlike ions from glass-salt equilibrium curves [56]. It must be underlined that all these approaches are based on equilibrium mixture models, even if it has been demonstrated that in most cases the ion exchanged glass layers reach the equilibrium structure after very long exchange times.

From the point of view of the inter-diffusion equation (6), this approach leads to the insertion in the inter-diffusion coefficient of two self-diffusion coefficients depending on the ion concentration in the following way:

$$\begin{aligned}D_A &= D_{A0} \exp\left(-\frac{X_A \epsilon_{AA}}{kT} - \frac{X_B \epsilon_{AB}}{kT}\right), \\ D_B &= D_{B0} \exp\left(-\frac{X_B \epsilon_{BB}}{kT} - \frac{X_A \epsilon_{BA}}{kT}\right)\end{aligned}\quad (8)$$

where it can be assumed that $\epsilon_{ij} = \epsilon_{ji}$. In the literature some empirical approaches can be found where a simple exponential dependence on the concentration is assumed [57].

A more complex approach, with relationships for self-diffusion coefficients like in Eq. (8), has been carried out by Gonella et al. in simulating the $\text{Cu}^+ - \text{Na}^+$ ion exchange process [38,58]. In this case, the presence of two Cu species diffusing into the glass under different oxidation states makes the description of the phenomenon much more complex. In particular, by integrating the coupled diffusion equations of the three migrating species, correlation effects between the charge fluxes have been evidenced, giving rise to the establishing of a Cu^+ concentration profile with a maximum below the surface. This result was confirmed by the analysis of XAFS spectra obtained at different depths, where the different $\text{Cu}^+/\text{Cu}^{2+}$ ratios have been calculated by supposing a mixture of CuO and Cu_2O like structures for the cation sites. Moreover, the experimental concentration profiles can be accurately reconstructed by expressing the self-diffusion coefficients with an exponential dependence on the concentration of the migrating species, multiplied by an interaction energy factor. The interaction energy values reported in Ref. [50] indicate a strong correlation between Cu^+ and Cu^{2+} ions with respect to the other species, even if this result requires further confirmation.

Finally, it has to be mentioned the approach carried on by Araujo [59,60] and Houde-Walter and co-workers [61–65]. In this case, the authors evaluated the dependence on the concentration of the logarithmic derivative of the ion activity into the glass using statistical thermodynamics relationships, leaving constant the self-diffusion coefficients. In the resulting formula the cation coordination, the ion-ion interaction energy and a phenomenological parameter take the role of fitting parameters. After fitting the diffusion coefficient calculated as a function

of the ion concentration by means of the Boltzmann–Matano method, the obtained parameters allow to reconstruct any concentration profile obtained in the same treatment conditions. The proposed model is very effective in reproducing the ion exchange process of a single ion and, moreover, outlines several interesting questions on the correlation between the ion exchange thermodynamics and the cation site microstructure. Nevertheless, it is lacking of fundamental considerations about the activation energy affecting the self-diffusion coefficients. In fact, the author themselves recognized that by using the available data on radioactive tracer diffusion measured in the corresponding glasses at different concentrations, one of the phenomenological parameters can be avoided [65].

To summarize this last section, it can be stated that ion exchange is a quite complex process, involving structural rearrangements of the ion sites and of the geometric placement of silica tetrahedra. Even if in most cases the resulting structure does not really reproduce the structure of mixed-alkali, the models developed for describing the MAE can be useful for reconstructing the inter-diffusion mechanism. But, in order to avoid the multiplication of adjustable parameters in the phenomenological coefficients, a more complete analysis of the microscopic and thermodynamic aspects of the glass network changes must be carried on. First of all, the problem has to be faced for a well-established glass compositions. In fact, experimental findings underline the role in the ion migration of the amount of NBOs, which depends also on the concentration of Al_2O_3 or B_2O_3 . Then, it is needed a deeper analysis of the meaning of the interaction energies and how they are related to the formation of bonds, of local rearrangements and of NBOs. Moreover, different analysis techniques can be used for evidencing the changes of the glass microstructure and of the cation sites.

5. Field-assisted ion exchange

The last argument to be underlined concerns the field-assisted ion exchange (FAIE). FAIE in molten salt baths has been used in the production of square diffusion profiles both for surface strengthening and for the production of optical waveguides with peculiar refractive index profiles [66,67]. This treatment procedure was also used for the production of buried waveguides or for tailoring the diffusion profiles, for example by inverting the applied field after the exchange [57,68,69].

From the point of view of the diffusion process, the equations describing the ion migration are the same outlined above, where the electric field is the sum of the local and the external fields, and a ion current crosses the glass slide. From the analytical point of view, the problem has been studied by Cooper at first [23,70] and then by Oven and co-workers [68,71,72], giving several approximate solutions both for single and multiple ion exchange treatments.

Due to operational setup difficulties when dealing with molten baths, most of the FAIE experiments were realized by promoting the ion migration into the glass from a metallic solid film deposited on the surface [73–77]. Two electric contacts, usually Al or Au, are deposited on the film and on the glass opposite face and connected by a voltage or a current generator.

The external electric field induces the oxidation of the metallic film and the electromigration of ions in the glass network, whereas Na^+ migrates to the opposite side.

The great advantage of this procedure is to largely extend the amount of ion species which can be introduced into the glass network. Since from the first electromigration experiments the most used element was silver, due to its wide application in the fabrication of passive optical devices and to the possibility of producing buried waveguides, by promoting the exhaustion of the whole film. In the last years, Cu^+ containing samples have been realized [14,78–80] and, more recently, Co^{2+} , Au^{3+} and Er^{3+} diffusion profiles into soda-lime glasses have been obtained by Gonella and co-workers [17,18,81,82].

In spite of the intriguing possibilities opened by FAIE from thin films, several technical problems related to the cation diffusion mechanism are still open. At first, space charge can occur near the ion-injecting anode if the diffusion rate of invasive ions is not equal to the rate at which the indigenous ions move to the cathode. The same problem can occur if the cathode cannot remove or neutralize the glass ions at the required rate. Moreover, the formation and diffusion of ions at the glass–film interface is a critical issue in controlling the process rate and the homogeneity of the sample. Up to now, few papers deal with the behavior of the glass–film interface during the FAIE. Among them, the deeper analysis has been presented by Kapila and Plawsky [75,77], who assumed the formation of an oxide buffer layer as ion source for the glass, describing the mechanism with the electrochemical cell hypothesis and describing the current behavior in terms of the film consumption. Actually, several problems are related to solid-state electrochemistry, like the presence of a critical electric field for triggering the ion diffusion, which is 3 V/mm for silver and around 50 V/mm for gold as observed by Cattaruzza et al. [82].

In the light of these considerations, also the inter-diffusion equations should be revised, by introducing the proper surface ion flux, as indicated by Kapila, and taking into account the development of space charges in the network when the ion mobilities are strongly different. In particular, local effects giving raise to the collapse of two NBOs to form a bridge should be accounted for, since the dramatic changes of the network structure, resembling that of electrically poled glasses, can heavily affect the ion migration.

6. Conclusions

In this paper, some experimental results have been described evidencing the change of the cation site during ion exchange in silicate glasses. These results point out the need of self-diffusion coefficient depending on the cation concentration to accurately describe the inter-diffusion mechanism and the resulting diffusion and concentration profiles. In particular, the rearrangement of the sites leads to the change of the diffusion activation energies, owing to the presence of terms describing the interaction of an ion species with its neighbors. The use of concentration-dependent activation energies allow the accurate reconstruction of diffusion profiles in quite complex situations, like those when three ion species diffuse

at the same time. The work in progress in this field is the study of the meaning of the interaction energies, in order to find their expression as a function of physical parameters describing the local structure and the rearrangement of the network.

References

- [1] D.K. Hale, *Nature* 217 (1968) 1115.
- [2] R.V. Ramaswamy, R. Srivastava, *J. Lightwave Technol.* 6 (1988) 984.
- [3] S. Honkanen, B.R. West, S. Yliniemi, P. Madasamy, M. Morrell, J. Auxier, A. Schulzgen, N. Peyghambarian, J. Carriere, J. Frantz, R. Kostuk, J. Castro, D. Geraghty, *Phys. Chem. Glasses—Eur. J. Glass Sci. Technol. Part B* 47 (2006) 110.
- [4] G. Jose, G. Sorbello, S. Taccheo, E. Cianci, V. Foglietti, P. Laporta, *J. Non-Cryst. Solids* 322 (2003) 256.
- [5] G.C. Righini, M. Brenci, A. Chiasera, M. Ferrari, M. Montagna, G. Nunzi Conti, S. Pelli, *Glass Sci. Technol.* 76-C2 (2003) 29.
- [6] V.M. Sglavo, D.J. Green, *J. Am. Ceram. Soc.* 84 (2001) 1832.
- [7] D.K. Tagantsev, G.O. Karapetyan, A.A. Lipovskii, V.V. Loboda, *J. Eur. Ceram. Soc.* 21 (2001) 2015.
- [8] M.I. Ojovan, A. Pankov, W.E. Lee, *J. Nucl. Mater.* 358 (2006) 57.
- [9] G.H. Chartier, P. Jaussaud, A.D. de Oliveira, O. Parriaux, *Electron. Lett.* 13 (1977) 763.
- [10] V. Neuman, O. Parriaux, L.M. Walpita, *Electron. Lett.* 15 (1979) 704.
- [11] Z.G. Zhou, X.M. Duan, *Opt. Commun.* 266 (2006) 129.
- [12] F. Gonella, F. Caccavale, A. Quaranta, A. Sambo, *J. Mod. Opt.* 45 (1998) 837.
- [13] J. Spirkova, P. Nebolova, I. Jirka, K. Mach, V. Perina, A. Mackova, G. Kuncova, *Fiber Integr. Opt.* 21 (2002) 63.
- [14] N. Valles-Villareal, A. Villalobos, H. Marquez, *J. Lightwave Technol.* 17 (1999) 606.
- [15] T. Izawa, H. Nakagome, *Appl. Phys. Lett.* 21 (1972) 584.
- [16] P. Tresnakova, H. Malichova, J. Spirkova, M. Mika, *J. Phys. Chem. Sol.* 68 (2007) 1276.
- [17] F. Gonella, P. Canton, E. Cattaruzza, A. Quaranta, C. Sada, A. Vomiero, *Mater. Sci. Eng. C* 26 (2006) 1087.
- [18] F. Gonella, E. Cattaruzza, A. Quaranta, S. Ali, N. Argiolas, C. Sada, *Solid State Ionics* 177 (2006) 3151.
- [19] R. Terai, R. Hayami, *J. Non-Cryst. Solids* 18 (1975) 217.
- [20] H. Kahnt, *J. Non-Cryst. Solids* 203 (1996) 225.
- [21] O.L. Anderson, D.A. Stuart, *J. Am. Ceram. Soc.* 37 (1954) 573.
- [22] A. Quaranta, F. Gonella, *J. Non-Cryst. Solids* 193 (1995) 334.
- [23] A.R.J. Cooper, *J. Non-Cryst. Solids* 14 (1974) 65.
- [24] R.H. Doremus, *J. Phys. Chem.* 68 (1964) 2212.
- [25] H.M. Garfinkel, *J. Phys. Chem.* 72 (1968) 4175.
- [26] F. Gonella, A. Quaranta, *Pure Appl. Opt.* 2 (1993) 405.
- [27] B. Pantchev, Z. Nikolov, *IEEE J. Quantum Electron.* 29 (1993) 2459.
- [28] G.N. Greaves, A. Fontaine, P. Lagarde, D. Raoux, S.J. Gurman, *Nature* 293 (1981) 611.
- [29] M. Dubiel, G. Mosel, *Jpn. J. Appl. Phys. Part 1* 33 (1994) 5892.
- [30] M. Dubiel, S. Brunsch, U. Kolb, D. Gutwerk, H. Bertagnolli, *J. Non-Cryst. Solids* 220 (1997) 30.
- [31] F. D'Acapito, F. Gonella, E. Cattaruzza, S. Pascarelli, P. Mazzoldi, S. Mobilio, *Nucl. Instrum. Methods B* 120 (1996) 110.
- [32] M. Dubiel, R. Schmitz, K. Scheerschmidt, H. Hofmeister, *J. Non-Cryst. Solids* 193 (1995) 632.
- [33] C. Maurizio, F. D'Acapito, C. Sada, G. Battaglin, E. Cattaruzza, F. Gonella, this conference, 2007.
- [34] J. Shen, D.J. Green, *J. Non-Cryst. Solids* 344 (2004) 79.
- [35] F. Gonella, A. Quaranta, *J. Mod. Opt.* 39 (1992) 1401.
- [36] F. Gonella, *Opt. Lett.* 17 (1992) 1667.
- [37] C. Maurizio, F. D'Acapito, M. Benfatto, S. Mobilio, E. Cattaruzza, F. Gonella, *Eur. Phys. J. B* 14 (2000) 211.
- [38] F. Gonella, A. Quaranta, E. Cattaruzza, S. Padovani, C. Sada, F. D'Acapito, C. Maurizio, *Comput. Mater. Sci.* 33 (2005) 31.
- [39] E. Borsella, F. Gonella, P. Mazzoldi, A. Quaranta, G. Battaglin, R. Polloni, *Chem. Phys. Lett.* 284 (1998) 429.
- [40] E. Borsella, G. De Marchi, F. Caccavale, F. Gonella, G. Mattei, P. Mazzoldi, G. Battaglin, A. Quaranta, A. Miotello, *J. Non-Cryst. Solids* 253 (1999) 261.
- [41] E. Borsella, G. Battaglin, M.A. Garcia, F. Gonella, P. Mazzoldi, R. Polloni, A. Quaranta, *Appl. Phys. A* 71 (2000) 125.
- [42] J. Lee, T. Yano, S. Shibata, M. Yamane, *J. Non-Cryst. Solids* 222 (1997) 120.
- [43] J. Lee, T. Yano, S. Shibata, M. Yamane, *J. Non-Cryst. Solids* 246 (1999) 83.
- [44] T. Yano, K. Azegami, S. Shibata, M. Yamane, *J. Non-Cryst. Solids* 222 (1997) 94.
- [45] T. Yano, T. Nagano, J. Lee, S. Shibata, M. Yamane, *J. Non-Cryst. Solids* 270 (2000) 163.
- [46] M.D. Ingram, J.E. Davidson, A.M. Coats, E.I. Kamitsos, J.A. Kapoutsis, *Glastechn. Ber.* 73 (2000) 89.
- [47] G.H. Frischat, R. Kirchmeyer, *J. Am. Ceram. Soc.* 56 (1973) 552.
- [48] A.K. Varshneya, *J. Am. Ceram. Soc.* 58 (1975) 196.
- [49] C. Matano, *Jpn. J. Phys.* 8 (1933) 109.
- [50] R. Kirchheim, *J. Non-Cryst. Solids* 272 (2000) 85.
- [51] R. Kirchheim, D. Paulmann, *J. Non-Cryst. Solids* 286 (2001) 210.
- [52] A. Bunde, M.D. Ingram, P. Maass, K.L. Ngai, *J. Phys. A* 24 (1991) L881.
- [53] M. Tomozawa, *J. Non-Cryst. Solids* 152 (1993) 59.
- [54] G.N. Greaves, S.J. Gurman, C.R.A. Catlow, A.V. Chadwick, S. Houde-Walter, C.M.B. Henderson, B.R. Dobson, *Phil. Mag. A* 64 (1991) 1059.
- [55] G.N. Greaves, K.L. Ngai, *Phys. Rev. B* 52 (1995) 6358.
- [56] R.J. Araujo, S. Likitvanichkul, Y. Thibault, D.C. Allan, *J. Non-Cryst. Solids* 318 (2003) 262.
- [57] S. Batchelor, R. Oven, D.G. Ashworth, *J. Phys. D: Appl. Phys.* 31 (1998) 390.
- [58] F. Gonella, A. Quaranta, S. Padovani, C. Sada, F. D'Acapito, C. Maurizio, G. Battaglin, E. Cattaruzza, *Appl. Phys. A* 81 (2005) 1065.
- [59] R.J. Araujo, *J. Non-Cryst. Solids* 152 (1993) 70.
- [60] R. Araujo, *J. Non-Cryst. Solids* 349 (2004) 230.
- [61] S.N. Houde-Walter, J.M. Inman, A.J. Dent, G.N. Greaves, *J. Phys. Chem.* 97 (1993) 9330.
- [62] J.M. Inman, S.N. Houde-Walter, G.N. Greaves, A.J. Dent, B.R. Dobson, *Jpn. J. Appl. Phys. Part 1* 32 (1993) 667.
- [63] J.M. Inman, J.L. Bentley, S.N. Houde-Walter, *J. Non-Cryst. Solids* 191 (1995) 209.
- [64] J.M. Inman, S.N. Houde-Walter, B.L. McIntyre, Z.M. Liao, R.S. Parker, V. Simmons, *J. Non-Cryst. Solids* 194 (1996) 85.
- [65] S.N. Houde-Walter, *Solid State Ionics* 105 (1998) 257.
- [66] G.H. Chartier, P. Jaussaud, A.D.D. Oliveira, O. Parriaux, *Electron. Lett.* 14 (1978) 132.
- [67] G. Chartier, P. Collier, A. Guez, P. Jaussaud, Y. Won, *Appl. Opt.* 19 (1980) 1092.
- [68] S. Batchelor, R. Oven, D.G. Ashworth, *Electron. Lett.* 32 (1996) 2082.
- [69] R. Oven, S. Batchelor, D.G. Ashworth, *J. Phys. D: Appl. Phys.* 32 (1999) 650.
- [70] M. Abou-el-eil, A.R. Cooper, *J. Am. Ceram. Soc.* 63 (1979) 390.
- [71] R. Oven, D.G. Ashworth, M.C. Page, *J. Phys. C: Condens. Matter* 4 (1992) 4089.
- [72] R. Oven, *J. Appl. Phys.* 101 (2007).
- [73] S. Honkanen, A. Tervonen, *J. Appl. Phys.* 63 (1988) 634.
- [74] S. Honkanen, S.I. Najafi, P. Poyhonen, G. Orce, W.J. Wang, J. Chrostowski, *Electron. Lett.* 27 (1991) 2167.
- [75] D. Kapila, J.L. Plawsky, *AIChE J.* 39 (1993) 1186.
- [76] C. Thevenin-Annequin, M. Levy, T. Pagnier, *Solid State Ionics* 80 (1995) 175.

- [77] D. Kapila, J.L. Plawsky, *Chem. Eng. Sci.* 50 (1995) 2589.
- [78] H. Marquez, D. Salazar, A. Villalobos, G. Paez, J.M. Rincon, *Appl. Opt.* 34 (1995) 5817.
- [79] D. Salazar, G. Paez, A. Villalobos, H. Marquez, J.M. Rincon, *Rev. Mex. Fis.* 41 (1995) 848.
- [80] R. Oven, M. Yin, P.A. Davies, *J. Phys. D: Appl. Phys.* 37 (2004) 2207.
- [81] E. Cattaruzza, F. Gonella, G. Peruzzo, A. Quaranta, C. Sada, E. Trave, *Mater. Sci. Eng. B* 146 (2008) 163.
- [82] E. Cattaruzza, C. Battaglin, F. Gonella, S. Ali, C. Sada, A. Quaranta, *Mater. Sci. Eng. B* 149 (2008) 195.